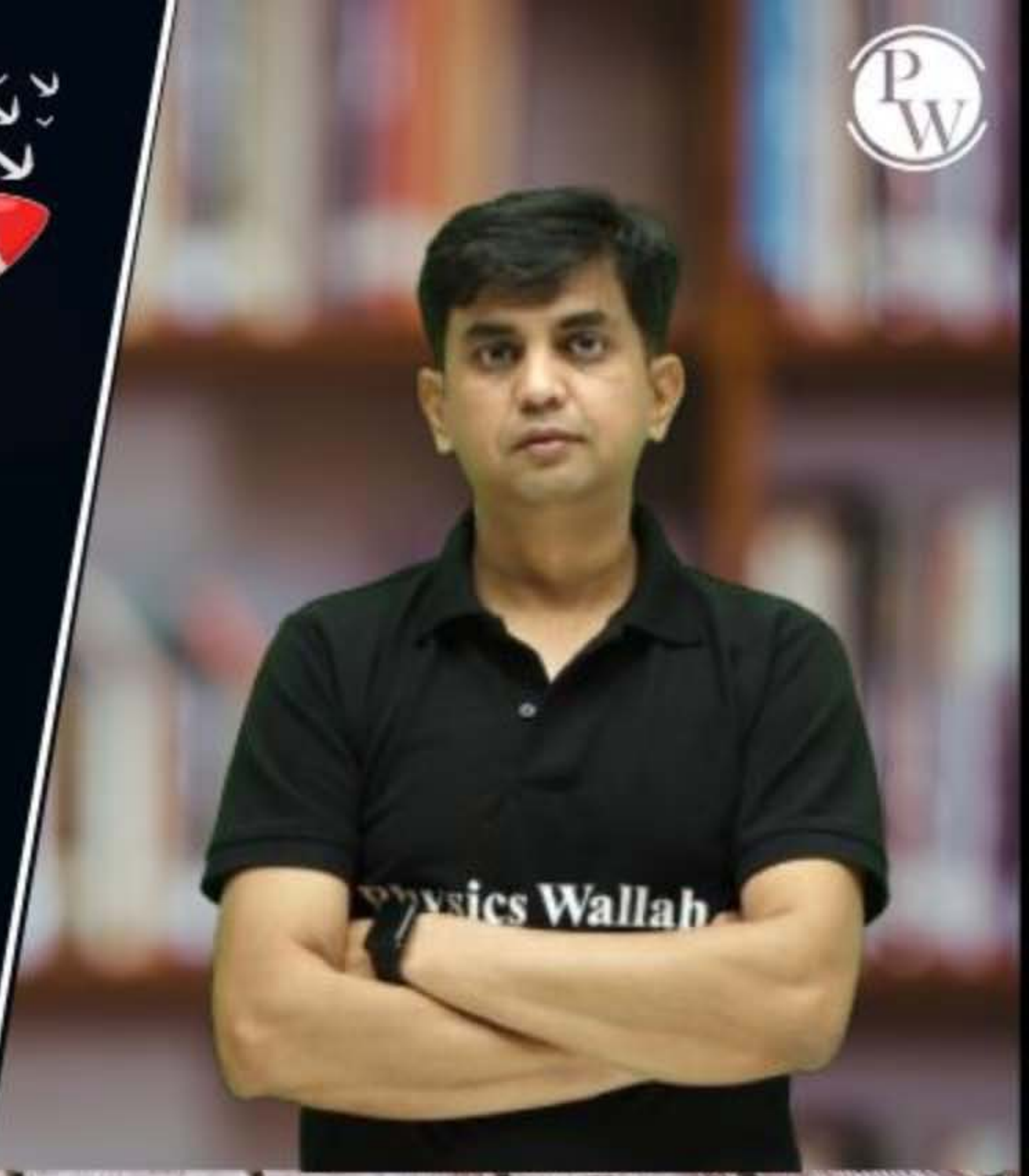
CS & IT ENGINEERING

Computer Network

IPv4 Address

Lecture No. - 02



By - Abhishek Sir

Recap of Previous Lecture

Topic Network Address

Topics to be Covered

Topic Broadcast Address

Topic Network Mask [Netmask]

ABOUT ME

Hello, I'm Abhishek

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Topic : Modes of Transmission

→ Three modes of transmission in IPv4 :

1. Unicast
2. Broadcast
3. Multicast

Topic : Unicast Transmission

→ One-to-one transmission

→ One host is sender and one host is receiver

Source IP Address = Host IP Address (of source host)

Destination IP Address = Host IP Address (of destination host)

Topic : Broadcast Transmission

→ One-to-All transmission

→ One host is sender and all host (of a network) are receiver

Source IP Address = Host IPv4 Address (of source host)

Destination IP Address = $\boxed{\text{Broadcast IPv4 Address}}$

Topic : Broadcast Address

→ Special IPv4 address (32 bits)

→ It can not be an IP address of any host in the network

→ Two types of IPv4 Broadcast Address :-

1. Limited Broadcast Address
2. Directed Broadcast Address

Topic : Limited Broadcast

→ Special IPv4 address (32 bits)

→ Used to broadcast a packet (to all hosts) of a network-segment [in which transmitting host belongs]

→ Limited Broadcast Address : 255 . 255 . 255 . 255

[All 32 bits are one]

$\boxed{11 - - - - - - - . 11 - - - - - - - . . 11}$

Topic : Directed Broadcast ✓ (By Default)

→ Special IPv4 address (32 bits)

→ Used to broadcast a packet (to all hosts) in a targeted network

→ Network Directed Broadcast Address ✓

NetID field = As Assigned

HostID field = All One Bits

Network Id	Host Id [1 1 1 1 1 . . . . 1 1]
x bit	y bit

Example 4 :-

[MCQ]

#Q. Consider a class-less IPv4 network where network id bits assigned to this network are "1100 1100 1010", identify correct decimal representation of broadcast address (special IPv4 address) which is used to broadcast a packet in this network ?

(A) 204 . 160 . 0 . 0

(B) 204 . 160 . 255 . 255

(C) 204 . 175 . 0 . 0

(D) 204 . 175 . 255 . 255

$\boxed{\text{Ans: D}}$

Network Directed Broadcast Address =

$$\underbrace{11001100\ 1010}_{\text{12 bit NetID}}\underbrace{1111\ 11111111\ 11111111}_{\text{20 bit HostID}}$$

204.175.255.255

Example 5 :- [MCQ]

#Q. Consider a class-less IPv4 network, the network address (special IPv4 address) of this network is "145 . 75 . 80 . 0" and size of network identifier field is 22 bits, which one of the following can be a broadcast address (special IPv4 address) of this network ?

(A) 145 . 75 . 80 . 0

(B) 145 . 75 . 80 . 255

(C) 145 . 75 . 83 . 255

(D) 145 . 75 . 95 . 255

$\boxed{\text{Ans: C}}$

Network Address = 145 . 75 . 80 . 0

$$\underbrace{10010001\ 01001011\ 010100}_{\text{22 bit Net ID}}\underbrace{00\ 00000000}_{\text{10 bit Host ID}}$$

Network Directed Broadcast Address =

$$\underbrace{10010001\ 01001011\ 010100}_{\text{22 bit Net ID}}\underbrace{11\ 11111111}_{\text{10 bit Host ID}}$$

145 . 75 . 83 . 255

Example 6 :- [H.W.] **[MCQ]**

#Q. Consider a class-less IPv4 network, the broadcast address (special IPv4 address) of this network is "175 . 175 . 175 . 255" and size of network identifier field is 22 bits, which one of the following can be a network address (special IPv4 address) of this network ?

(A) 175 . 175 . 0 . 0

(B) 175 . 175 . 160 . 0

(C) 175 . 175 . 172 . 0

(D) 175 . 175 . 175 . 0

Topic : Multicast Transmission

$\boxed{\text{IGMP}}$ ✓

→ One-to-many transmission

→ belongs to same same

→ One host is sender and group of hosts are receiver

Source IP Address = Host IP Address (of sending host)

Destination IP Address = Multicast Address

[Special IP Address as a Group ID]

* Class D IP Addresses

Topic : This Host

→ Special IP address (all zero bits)

→ Unspecified IP Address

→ This Host : 0 . 0 . 0 . 0

$\boxed{00 \ldots\ldots 00 \ldots\ldots 00}$

Topic : Host IP Address

→ Host IP address

→ Used to identify a host uniquely world wide

NetID field (x bits) = As Assigned

HostID field (y bits) = $\boxed{\text{Any thing}}$

[Except all zero and all one bits]

Example 7 :-

#Q. Consider a class-less IPv4 network where network id bits assigned to this network are “1100 1100 1010”, what is the first and last IP address of this network which can be assigned to a host ?

Network Address =

$$\underbrace{11001100\ 1010}_{\text{12 bit NetID}}\underbrace{0000\ 00000000\ 00000000}_{\text{20 bit HostID}}$$

First IP Address that can be assigned to a host =

$$\boxed{\underbrace{11001100\ 1010}_{\text{12 bit NetID}}\underbrace{0000\ 00000000\ 00000001}_{\text{20 bit HostID}}}$$

Last IP Address that can be assigned to a host =

$$\boxed{\underbrace{11001100\ 1010}_{\text{12 bit NetID}}\underbrace{1111\ 11111111\ 11111110}_{\text{20 bit HostID}}}$$

$(2^{20}-2)$ host IP

Network Directed Broadcast Address =

$$\underbrace{11001100\ 1010}_{\text{12 bit NetID}}\underbrace{1111\ 11111111\ 11111111}_{\text{20 bit HostID}}$$

Network Address = 204.160.0.0

First IP Address [that can be assigned to a host] = 204.160.0.1

Last IP Address [that can be assigned to a host] = 204.175.255.254

Network Directed Broadcast Address = 204.175.255.255

Example 8 :-

#Q. Consider an IPv4 address of a class-less IPv4 network is "115 . 120 . 125 . 130 / 19", write network address, broadcast address, first and last IP address of this network which can be assigned to a host ?

IPv4 Address = 115 . 120 . 125 . 130 / 19

= $\underbrace{115 . 120 . 011}_{19 \text{ bit NetID}}\underbrace{11101 \ \ 100000010}_{13 \text{ bit HostID}}$

Network Address	= 115.120.96.0
First IP Address [that can be assigned to a host]	= 115.120.96.1
Last IP Address [that can be assigned to a host]	= 115.120.127.254
Network Directed Broadcast Address	= 115.120.127.255

Network Address =

$$\underbrace{115\,.\,120\,.\,011}_{\text{19 bit NetID}}\underbrace{00000\ \ 00000000}_{\text{13 bit HostID}}$$

First IP Address that can be assigned to a host =

$$\underbrace{115\,.\,120\,.\,011}_{\text{19 bit NetID}}\underbrace{00000\ \ 00000001}_{\text{13 bit HostID}}$$

Last IP Address that can be assigned to a host =

$$\underbrace{115\,.\,120\,.\,011}_{\text{19 bit NetID}}\underbrace{11111\ \ 11111110}_{\text{13 bit HostID}}$$

$(2^{13} - 2)$ Host IP

Network Directed Broadcast Address =

$$\underbrace{115\,.\,120\,.\,011}_{\text{19 bit NetID}}\underbrace{11111\ \ 11111111}_{\text{13 bit HostID}}$$

[NAT] [H.W.] [GATE-2015][2 Mark]

#Q. In the network "200 . 10 . 11 . 144 / 27", the fourth octet (in decimal) of the last IP address of the network which can be assigned to a host is ______.

Topic : Network Size

→ Network Size : Maximum number of hosts can be in a network

→ HostID field (y bits)

$$\boxed{\text{Network Size} = [2^y - 2] \text{ hosts per network}}$$

Example 9 :- [NAT]

#Q. Consider a class-less IPv4 network, the number of bits in the network id field are 22, how many maximum number of hosts can be in this network?

Number of bits in Network ID field (x) = 22

Number of bits in Host ID field (y) = (32 – x) = 10

Maximum number of hosts can be in a network = $[2^y - 2]$ hosts per network

$= (2^{10} - 2)$ hosts per network

$= 1022$ hosts per network

$\boxed{\text{Ans} = 1022}$

2 mins Summary

Topic Broadcast Address

Topic ~~Network Mask~~ Network Size

THANK - YOU

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